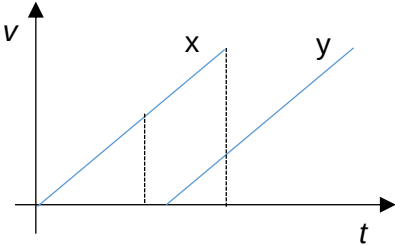


Qn	Key	guide
1.	A	Considering units of the equation, we have $s^2 = \frac{\text{kg m}^2}{\text{units of K}}$, so units of K = $\text{kg m}^2 \text{s}^{-2} = (\text{kg m s}^{-2}) \text{m} = \text{N m}$
2.	D	 Since air resistance is negligible, the gradient of both graphs will be the same. From the v-t graph, it can be seen that at any time, the difference in the area under graph for X and Y will be increasing hence the separation will be increasing.
3	B	Note that the carriages and the engine have a common acceleration, say a.